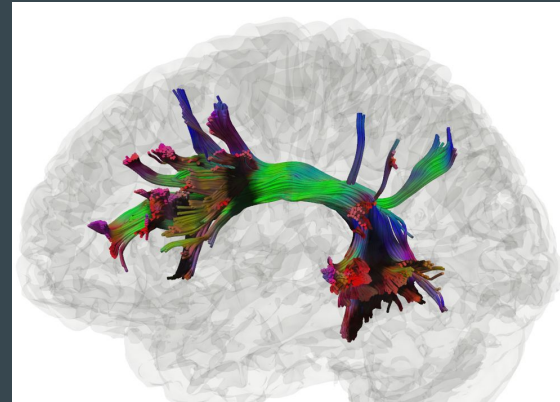


Evolution of the MRI

...

Joseph and Elizabeth



Overview: Stages in MRI Development

1

Early MRI Techniques

3

HARDI Method

2

DWI & DTI Methods

4

Q-Ball Imaging

Important Terminology

Voxel - “volumetric” + “pixel” (essentially a 3D pixel)

Brain Fibers - web of neural connections

Association Fibers - white matter, connections within same hemisphere of brain

Tractography - technique for 3D modeling white matter (bundles of nerve fiber)

Anisotropy - non-uniformity in different directions

Gradient Vector - gives the direction of neural fibers

*compare: direction and rate of fastest increase of a scalar field

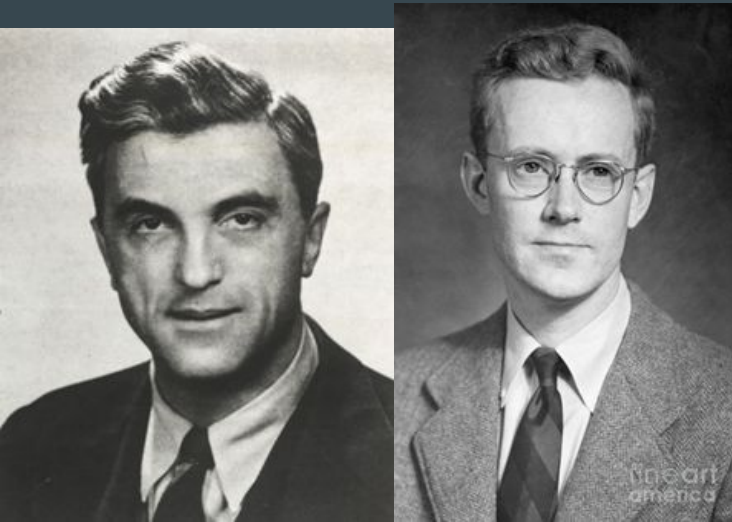
K-space - temporary “image space”; a matrix that holds the data collected during the MRI scan with the same number of rows and columns as the final image (often 2D)

*the domain of a fourier transform that maps data onto image

Earliest MRI Methods - Nuclear Magnetic Resonance (NMR)

1946 Physicists **Felix Bloch** and **Edward Purcell**
Nuclear magnetic resonance (NMR)

awarded the 1952 Nobel Prize in Physics.



1973 Physicist **Paul Lauterbur**
the first spatial localization MRI technique



1971 Scientist **Raymond Damadian**
different types of tissues
differences in magnetic resonance signals



Diffusion Weighted Imaging (DWI) and Diffusion Tensor Imaging (DTI)

1. Concept
2. Diffusion Type and Data Collection
3. Mathematical Model and Key Metrics
4. Limitations and Applications

DWI Math

$$S = S_0 e^{-bD}$$

Diffusion-weighted MRI (DWI) is based on the **Stejskal-Tanner equation**, which describes how MRI signal is affected by water molecule diffusion:

S : Signal intensity in the diffusion-weighted image

S_0 : Signal intensity without diffusion weighting ($b = 0$)

b : Diffusion sensitivity factor

$$b = \gamma^2 G^2 \delta^2 (\Delta - \delta/3)$$

D : **Apparent Diffusion Coefficient (ADC)**, which quantifies water diffusion per voxel

$$D = -\frac{\ln(S/S_0)}{b}$$

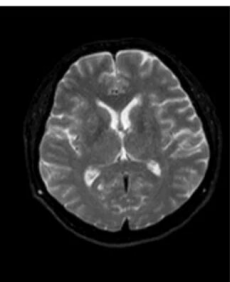
γ : Gyromagnetic ratio of protons (hydrogen nuclei)

G : Diffusion gradient strength

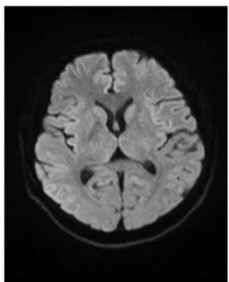
δ : Duration of diffusion gradient

Δ : Time interval between diffusion gradients

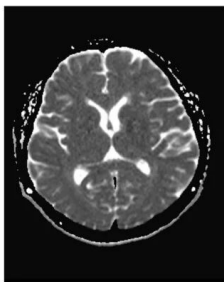
Diffusion-weighted imaging (DWI) of the Brain



b-value : 0



b-value : 1000



ADC MAP

DTI Math

$$S_i = S_0 e^{-b \mathbf{g}_i^T \mathbf{D} \mathbf{g}_i}$$

DTI extends DWI by using a **tensor model** to describe diffusion anisotropy. The fundamental equation is:

S_i : Signal intensity in the diffusion-weighted image along the i th gradient direction

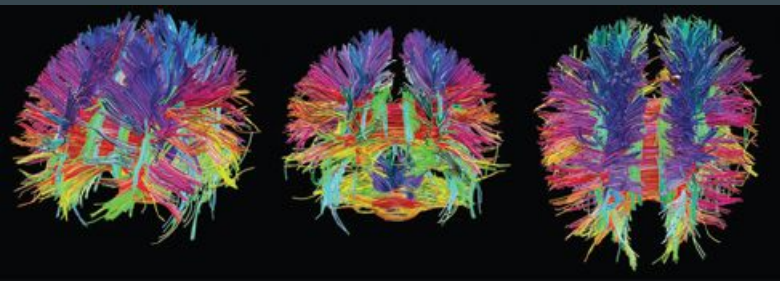
S_0 : Baseline signal intensity ($b = 0$)

b : Diffusion weighting factor

$\mathbf{g}_i$: Unit gradient direction vector (3×1)

$\mathbf{D}$: Diffusion tensor, a symmetric 3×3 matrix that describes diffusion in different directions:

$$\mathbf{D} = \begin{bmatrix} D_{xx} & D_{xy} & D_{xz} \\ D_{yx} & D_{yy} & D_{yz} \\ D_{zx} & D_{zy} & D_{zz} \end{bmatrix}$$



$$\mathbf{D} \mathbf{v}_i = \lambda_i \mathbf{v}_i$$

Eigenvalue decomposition is required to analyze diffusion characteristics..

Key Metrics:

In DWI:

$$S = S_0 e^{-bD}$$

$$D = -\frac{\ln(S/S_0)}{b}$$

Apparent Diffusion Coefficient (ADC): is a crucial parameter derived from DWI, quantifying the average diffusion rate of water molecules within a voxel.

In DTI:

$$S_i = S_0 e^{-b \mathbf{g}_i^T \mathbf{D} \mathbf{g}_i}$$

$$FA = \sqrt{\frac{3}{2}} \cdot \frac{\sqrt{(\lambda_1 - MD)^2 + (\lambda_2 - MD)^2 + (\lambda_3 - MD)^2}}{\sqrt{\lambda_1^2 + \lambda_2^2 + \lambda_3^2}}$$

Fractional Anisotropy (FA): Measures diffusion directionality, ranging from 0 (isotropic, uniform diffusion) to 1 (fully anisotropic, one-directional diffusion).

Mean Diffusivity (MD): Quantifies overall diffusion magnitude, related to tissue integrity.

Axial Diffusivity (AD) and Radial Diffusivity (RD): Used to assess axon and myelin integrity in neurons.

$$\mathbf{D} \mathbf{v}_i = \lambda_i \mathbf{v}_i$$

$$MD = \frac{\lambda_1 + \lambda_2 + \lambda_3}{3}$$

$$AD = \lambda_1$$

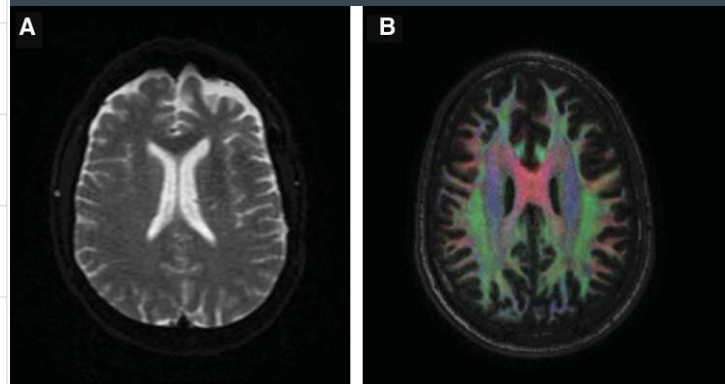
$$RD = \frac{\lambda_2 + \lambda_3}{2}$$

Eigenvalues λ , representing diffusion rates along the principal axes

Eigenvectors $\mathbf{v}$, indicating the corresponding diffusion directions

Diffusion Weighted Imaging (DWI) vs. Diffusion Tensor Imaging (DTI)

Feature	Pure Diffusion Model (DWI)	Diffusion Tensor Model (DTI)
Diffusion Type	Assumes isotropic diffusion	Accounts for anisotropic diffusion
Data Collection	Measures diffusion in a single direction	Measures diffusion in multiple directions (≥ 6 , typically >20)
Mathematical Model	Mono-exponential decay equation	3×3 diffusion tensor matrix
Output Parameter	ADC (Apparent Diffusion Coefficient)	FA (Fractional Anisotropy), MD (Mean Diffusivity), AD, RD
Structural Information	✗ Cannot resolve fiber orientation	✓ Can reconstruct white matter tracts
Limitations	No directional information, unsuitable for fiber tracking	Cannot resolve crossing fibers , limited angular resolution
Suitable for	Stroke detection, tumor differentiation	Brain connectivity, neurodegenerative diseases



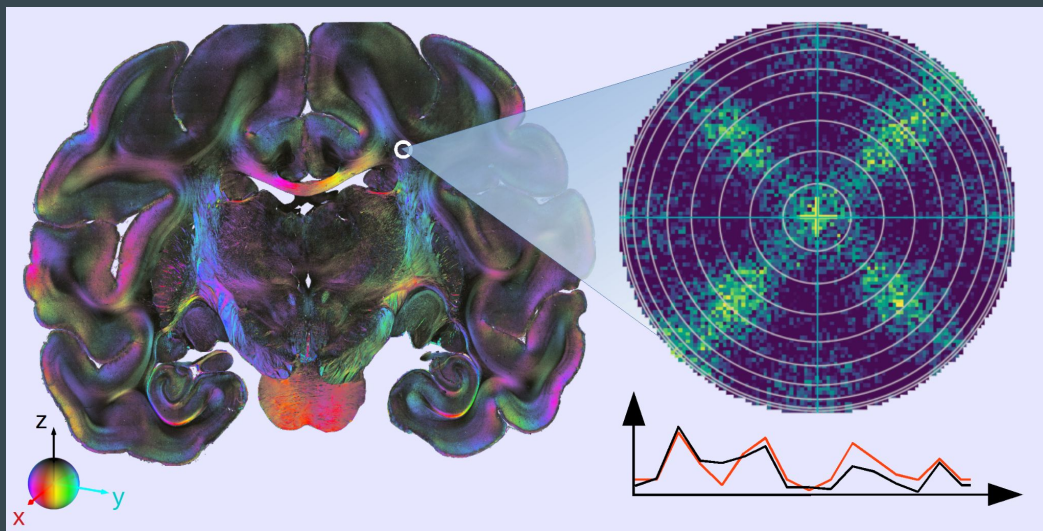
Images from Source 8.

generated with ChatGPT

DWI can only measure the **total diffusion magnitude** and does **not** provide **directional information** ;
DTI obtains diffusion **direction and anisotropy information** through eigenvalue decomposition, making it suitable for white matter fiber tracking, but it **cannot resolve crossing fibers**.

Shortcomings of Diffusion Tensor Imaging

crossing fibers !!



“can only resolve a single fiber direction within each voxel” (Source2.)

**90% of voxels contain crossing fibers!* (Source 3)

simplistic & misleading

Green (Anterior-Posterior)
Blue (Superior-Inferior)
Red (Left-Right)

What is HARDI?

High Angular Resolution Diffusion Imaging

HARDI $\neq$ interpolation or data analysis method

HARDI = a means of collecting data

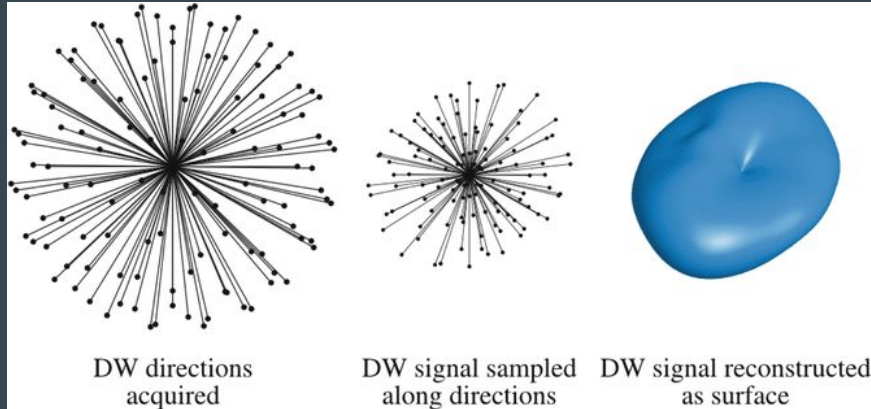


Image from Source 1.

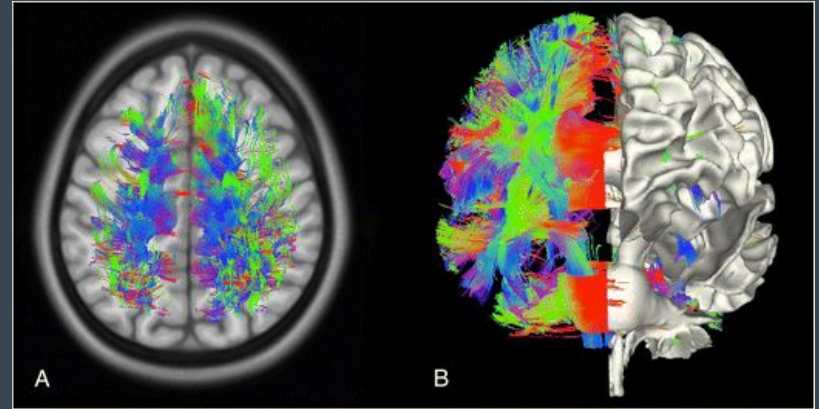
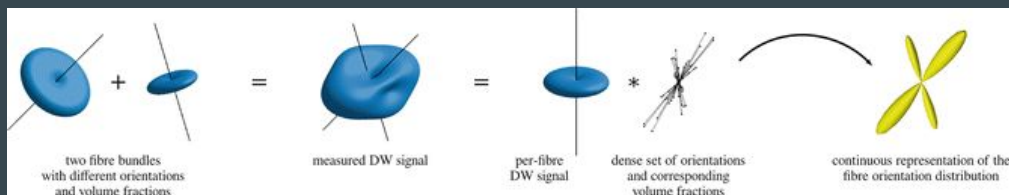


Image from Source 4.

Goal of HARDI (High Angular Resolution Diffusion Imaging)

isolate gradient vectors contributed by two crossing fibers within a voxel



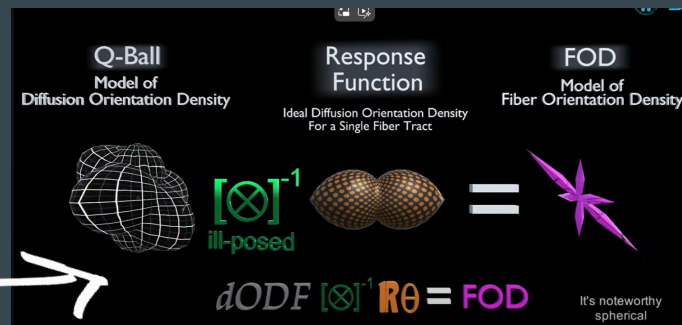
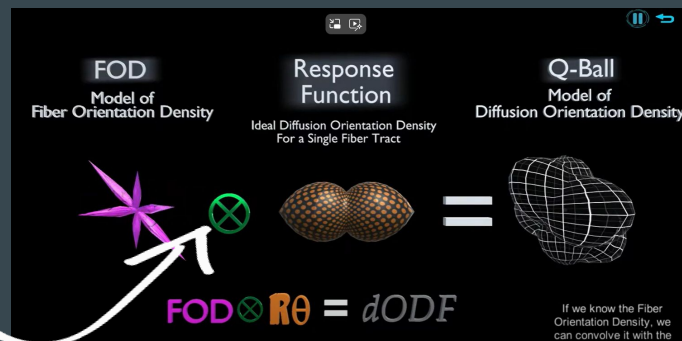
measured DW signal in a voxel (sum of all measured DW signals)



decompose into distinct fiber orientations

“Convolve” = blending two functions, allowing the shape of one to impact the other

“Deconvolution”



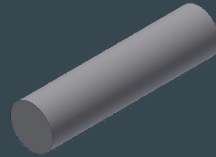
Shortcomings of HARDI

DW Signal



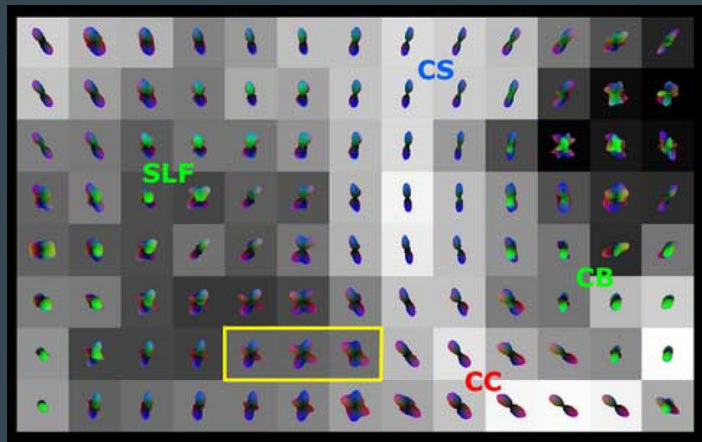
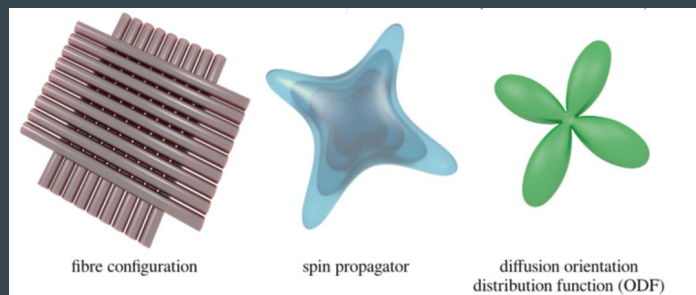
assumptions and approximations:

the makeup of white matter fibers



Tractography

Q-Ball Imaging Model-Free Methods



Characterizing **diffusion** rather than **fiber configuration**

1. Spin displacement probability density function

$$P(r|x,\Delta)$$

x = initial position | r = displacement | Δ = diffusion time

2. Diffusion orientation density function

Probability that a molecule will move in a given direction
(no data about displacement)

*less complicated $\rightarrow$ efficient, and it's adequate!

QUESTIONS?



Sources

1. [High Angular Resolution Diffusion Imaging | Radiology Key](#)
2. [Q-ball imaging](#)
3. [High angular resolution diffusion-weighted imaging \(HARDI\): Beyond Diffusion Tensor Imaging](#)
4. [a 3D depiction of whole brain tractography obtained with HARDI of an... | Download Scientific Diagram](#)
5. [MRI Illustrated: Diffusion, Hardi & Q-Balls](#)
6. [Clinical feasibility of using mean apparent propagator \(MAP\) MRI to characterize brain tissue microstructure - PMC](#)
7. [Development and Initial Evaluation of 7 Tesla Q-ball Imaging of the Human Brain](#)
8. [Michael & Ware, Marcus. \(2018\). Advances in Neuro-Oncology Imaging Techniques](#)
9. [Magnetic resonance imaging](#)
10. [Diffusion-Weighted Magnetic Resonance Imaging](#)
11. [Diffusion Tensor Imaging](#)



Diffusion Weighted Imaging (DWI) vs. Diffusion Tensor Imaging (DTI)

